

- 6 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{1}{8} & 0 \leq x < 1, \\ \frac{1}{28}(8-x) & 1 \leq x \leq 8, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the cumulative distribution function of X . [3]

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- (b) Find the value of the constant a such that $P(X \leq a) = \frac{5}{7}$. [3]

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The random variable Y is given by $Y = \sqrt[3]{X}$.

- (c) Find the probability density function of Y . [5]

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